

Remote sensing in rangeland fire ecology: Comparing imagery to measured fire behavior, and burn severity across prescribed burns and wildfires

Response to review

AR: Author Responses (AR) appear below in this green text.

Academic editor

We received comments from 3 reviewers, all of whom provided good suggestions to improve the manuscript. Please address all review comments provided and respond with a detailed point-by-point response. Please note that one of the reviewers encouraged the use of a combined results and discussion section. The guide to authors lists the following as the recommended format of a manuscript: abstract, keywords, introduction, materials and methods, results, discussion, and conclusions (optional) sections. From this perspective, I encourage you to keep results and discussions separate. However, the main issue raised by the reviewer regarding more discussion around mechanisms is valid and needs to be discussed (reviewer 2 also raises this as an issue). It is also worth noting that the reviewers all commented on additional analysis.

AR: I appreciate the opportunity to consider these thoughtful reviews and revise accordingly. Major changes to this version include:

1. *Higher-quality data with stronger trends.* While not identified by reviewers, since submission I learned of an important step in scene normalization that corrects timeseries imagery for potentially confounding sources of error among scenes. I completely re-processed the remotely-sensed data accordingly and worked in additional statistical tests following reviewer comments. No trends in the data changed and thus nothing in the interpretation and conclusions have been affected, although a couple statistical results were more robust and thus provide better discussion points.
2. *Summary tables.* Two reviewers requested data summaries and these have been added to the Appendix.
3. *More discussion of mechanisms.* Several paragraphs have been added to the Discussion and other relevant ones moved up to ensure a thorough discussion of the trends observed and the processes underlying them prior to the broader discussion of management implications.